15 Environmental

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Topic Highlights

Risk Reduction
Economic Incentives
Building Controls
Environmental Control Issues

15.1 Introduction

Environmental controls are installed for a variety of reasons and incentives. They are like the traditional "carrot and stick" approach to modify behavior. There are significant rewards (carrots) possible as a result of environmental controls, but there are also significant penalties (sticks) that can result from poor implementation of environmental controls.

15.2 Risk Reduction

Environmental controls are one way of limiting liability and penalties such as fines and, potentially, incarceration. Environmental monitoring such as a Continuous Emission Monitoring System (CEMS) and other environmental systems for monitoring controlled and fugitive emissions from a facility are the most common way of ensuring a facility is in compliance with the associated operating licenses and regulations. In the United States, it is the responsibility of the Environmental Protection Agency (EPA) to mandate, through the appropriate Code of Federal Regulations (CFR), the rules and regulations regarding acceptable limits and measurement methods for a variety of substances considered harmful to the environment. In many cases, these operating limits form part of the license to operate a facility, meaning it is possible that lack of compliance with the regulations stipulated in the license can result in the temporary or permanent closure of a facility.

Leak Detection and Repair (LDAR) programs are another legislated tool used to minimize fugitive emissions from a facility. This program is not just a "stick" since it comes with the "carrot" that, if a facility consistently demonstrates low levels of fugitive emissions, the frequency at which the potential leak points need to be monitored can be reduced. A consistently "clean" environmental record not only reduces the potential of the "stick," but it is also the first of many "carrots." The broader community in which a company operates recognizes environmental efforts as those of a good corporate citizen adding value to a region through employment opportunities with negligible negative side effects.

15.3 Economic Incentives

Many companies have found that by tracking and minimizing their losses to the environment of a variety of substances, including their desired product, they are able to improve the operation of their

facility. Obviously, every unit of product not lost to the environment results in increased revenue, because it is now being sold as product. In addition, facilities are finding that, by concentrating their effluent stream, it is possible to sell the resulting concentrate as a specialty product or, alternately, as a feed stock to a nearby facility to convert to a higher value product of their own.

The economics, or feasibility, of each recovery project can only be determined locally. However, what is "given" is that you cannot control what you cannot measure, so proper measurement and control of the process, and especially the stream constituents of interest, is critical. Fortunately, the analytical technology that is under development and becoming available continues to improve in reliability, repeatability, sensitivity, and accuracy.

What is commonly accepted and has been demonstrated is that converting a process loop from open to closed loop control results in a halving of the standard deviation of the system. This is demonstrated in Figure 15-1 that shows how a process will continue to have the same number of incidents when it will exceed the operating constraint but because the standard deviation is less, it is possible to move the mean, representing the process setpoint closer to that limit.

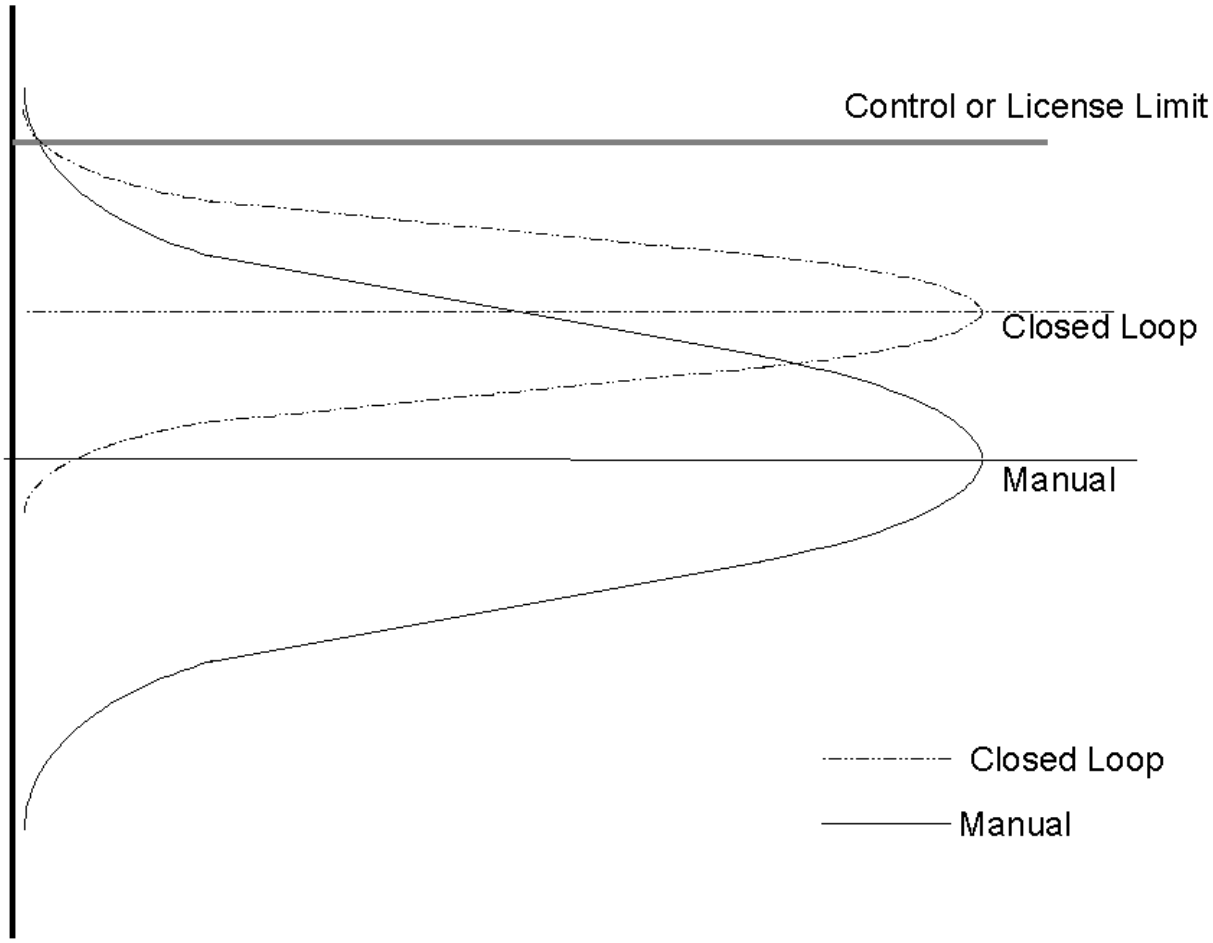


Figure 15-1: Impact of Tighter Loop Control

15.4 Building Controls

An often overlooked component of environment controls is the control of buildings in which workers and, in many cases, processes are housed.

In the case of workers, it is important to maintain a comfortable environment with a limited range of temperature and fresh air. "Sick Building Syndrome" was a periodic problem in the past because—in the quest for energy efficiency that included ensuring a building was "air tight"—a number of facilities were built that did not have enough fresh air being exchanged to prevent the concentration of gases, such as carbon dioxide and vapors from the building components themselves. This resulted in people having headaches and other symptoms that negatively impacted their ability to work.

Building controls are also taking on a more important role in industries as well, especially in the pharmaceutical and electronics industries. Both types of facilities have requirements for "clean environments" and, in the case of pharmaceutical facilities, the temperature must be maintained within a limited range. Electronics, of course, also require maintenance of building humidity, as well as a means to control static electricity.

For this reason, many facilities now have control systems dedicated to maintenance of the building environment/envelope that are comparable to the ones used for plant automation, often sharing the same components used to control the operating facility itself.

15.5 Environmental Control Issues

Environmental controls share many commonalities with other control systems and components of a facility. In addition, many analyzers used for detecting environmental constituents require regular maintenance to replenish consumables and maintain/verify the accuracy of measurements required of them by legislation.

15.5.1 Accuracy

In most control systems, repeatability is more important than accuracy. However, in the case of environmental monitoring, accuracy is also important. Accuracy of environmental systems, as verified by calibration and/or reference checking, must include the entire system from sample probe through to the signal, as received by the control system.

A Statistical Quality Control (SQC) process should be used to monitor any change, with the introduction of each reference gas sample and a change to the system's calibration factor made *only* when it is statistically significant *and* the reason for the change has been identified and resolved. Doing otherwise will result in the system continuing to "chase" the calibration gas and will not improve system accuracy at all.

15.5.2 Maintenance

All equipment and software applications require maintenance; environmental systems are no different. Without maintenance, the equipment will deteriorate and cease to function. Because of the specialized nature of the equipment used for environmental measurement, especially analyzers, the most effective way of ensuring continued reliable signals is to have a dedicated support team in place. This team should be made up of maintenance technicians/mechanics, engineers/technicians, and chemists working together to keep systems operating as they should. A poorly maintained system is a liability to a company, since it costs money to install and does not provide the benefit used to offset this cost.

15.6 References

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3. ISA SP71 Series of Standards. *Environmental Conditions for Process Measurement and Control Systems.*

About the Author

Ian Verhappen is an ISA Fellow and Director at ICE-Pros, Inc. an independent instrument and control engineering consulting firm specializing in process analyzer systems, fieldbus, and oil sands automation. He is a past ISA Analysis Division director and has been editor of the division newsletter since 1990. He has also contributed to the *Analytical Instrumentation* volume of ISA's Practical Guides for Measurement and Control Series, as well as the most recent editions of the *Instrument Engineer's Handbook*, jointly published by ISA and CRC Press.